

Selena (Ruiqi) Ge

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EDUCATION

University of California, San Diego, MS in Data Science Sep 2024 - May 2026

- Cumulative GPA: 4.0/4.0

New York University, Shanghai, BS in Mathematics Sep 2020 – May 2024

- Major GPA: 3.95/4.0
- Minor: Business and Finance
- **Honors:** Major Honors in Arts and Sciences(2 in Math discipline), NYU Founder's Day Award, Latin Honors: Cum laude, NYU Shanghai Excellence Award, Dean's list for 4 academic years.

RESEARCH INTERESTS

Diffusion Models, Machine Learning(ML), Reinforcement Learning(RL), Computer Vision(CV), Multimodal Models, Stochastic Processes, Causality.

PUBLICATIONS & SYMPOSIUMS

Fan Feng*, **Selena Ge***, Minghao Fu*, et. "Ada-Diffuser: Latent-Aware Adaptive Diffusion for Decision-Making." *Preprint. Under Review for NeurIPS 2025.*

Ruiqi Ge*, Jiaqi Liu*. "Bayesian Modeling of Information Spillovers in Post-Pandemic Travel Decisions ."

- **Oral Presentation** in *The 18th International Research Symposium on Service Excellence in Management (QUIS18)*, 2023.
- **Oral & Poster Presentation** in *NYU Dean's Undergraduate Research Fund (DURF) Symposium*, 2022.

SELECTED RESEARCH

Hierarchical Sparse Autoencoder(SAE) for Controllable Image Editing May 2025 – Present
Lead | Advisor: Prof. Biwei Huang

- Investigate limitations in current Matryoshka Sparse Autoencoder (MSAE) approaches, identifying critical gaps in validation of hierarchical concept learning and formulating targeted research objectives to enhance controllable image editing.
- Conduct empirical validation across diverse image domains using statistical dependence testing and hierarchical clustering analysis to quantitatively demonstrate emergence of multi-level semantic concepts.

Controllable Image Generation via Compact Representations May 2025 – Present
Lead | Advisor: Prof. Biwei Huang

- Propose and implement a generation framework addressing the entanglement problem in visual representation learning, combining self-supervised features (DINO) with explicit structural decomposition.
- Design a compression strategy using autoencoder(AE), reducing computational complexity from $O(n^6)$ to $O(n^4)$ for faster inference while maintaining semantic fidelity through 768D $\rightarrow$ 64D feature compression.

Ada-Diffuser: A Causal Diffusion Framework for Latent Identification in RL Mar 2025 – May 2025
Co-First | Advisor: Prof. Biwei Huang

- Contributed to the theoretical foundation of Ada-Diffuser, a novel framework that integrates latent variable identification with causal diffusion modeling for reinforcement learning.
- Derived identifiability conditions under minimal temporal observations and supported the formulation of latent-contextual Partially observable Markov decision process(POMDPs) using structural causal models (SCMs).
- Provided key insights on integrating diffusion models into sequential decision-making by modeling autoregressive causal processes with time-dependent latent factors.

Enhance Cross-Cultural Toxic Speech Detection with Concept Bottleneck LLMs

Oct 2024 – Jan 2025

Independent Study | Advisor: Prof. Lily Weng

- Initiated the idea of extending Concept Bottleneck LLMs (CB-LLM) to multilingual toxic speech detection, targeting fairness and interpretability across cultural and linguistic contexts.
- Formulated the theoretical foundation for a four-stage pipeline (concept generation, automatic scoring, bottleneck alignment, and prediction), leveraging SimCSE and ACS for cultural concept grounding.
- Contributed to the design of experiments and evaluation strategy, helping guide implementation efforts and ensure alignment with ethical and explainable AI principles in real-world moderation tasks.

Stochastic Differential Equations(SDE) for Large-scale Machine Learning

Mar 2023 – Jan 2024

Undergraduate Senior Thesis | Advisor: Prof. Mathieu Lauriere & Prof. Roberto Fernandez

- Clarified the connection between SDE and two cutting-edged diffusion models, demonstrating how score-based methods can reverse the noise addition process inherent in diffusion models.
- Carried out theoretical foundations and numerical experiments to confirm the efficacy of Denoising Diffusion Probability Model(DDPM) and Score Matching Langevin Dynamics (SMLD) within the SDE framework.

Modeling Stochastic Gradient Descent(SGD) with Continuous-Time Dynamics

Mar 2023 – Jan 2024

Independent Study | Advisor: Prof. Loucas Pillaud-Vivien & Prof. Roberto Fernandez

- Mathematically proved that SGD learning dynamic is equivalent to a multi-dimensional Ornstein-Uhlenbeck process by modeling the discrete iterative process as a continuous-time SDE.
- Validated the theoretical model through numerical simulations, implementing both the SGD algorithm and the corresponding SDE model via the Euler-Maruyama method.

Bayesian Modeling of Information Spillovers in Post-Pandemic Travel Decisions

Jan 2022 – Jun 2023

Co-First | Advisor: Prof. Jiding Zhang Fernandez

- Investigated the impact of virtual tours on post-pandemic tourism by developing a Bayesian model to quantify how consumers update travel intentions based on potentially biased influencer signals.
- Engineered a web crawler to acquire over 12,000 data points from Bilibili, processed and integrated the raw data with official government reports to construct a 5-year panel dataset for 6 major cities.

TEACHING EXPERIENCE

Lead Learning Assistant at New York University(NYU) & NYU Shanghai

Sep 2021 – May 2022

Linear Algebra(MATH-SHU 140) - Spring 2022 - NYU Shanghai

Pre-Calculus(MATH-SHU 9) - Fall 2021 - NYU Shanghai

Calculus I(MATH-UA 140) - Fall 2021 - NYU (remote)

- Organized academic workshops, learning assistant training sessions, and discipline-related projects at NYU Shanghai Academic Resource Center; received Learning Assistant Excellent Communication Award.
- Hosted biweekly review sessions for over 60 students on *Linear Algebra* and *Calculus* to enhance their comprehension of new concepts and provided after-class tutoring for 4 hours every week.

PERSONAL SKILLS

Libraries: PyTorch, TensorFlow/Keras, NumPy, Scikit-Learn, Matplotlib, Pillow, OpenCV

Developer Tools: Git, Docker, HTML, Weights & Biases, LaTeX, LLM APIs

Hobby: Chinese Folk dancing, Figure Skating, Creative Writing, and Playing with my Cat.